

## Applications



- 1 An investment scheme advertises the following percentage returns after 2 years based on historical probabilities:

|                           |     |      |      |
|---------------------------|-----|------|------|
| Percentage return ( $x$ ) | 10% | 15%  | 25%  |
| $P(X = x)$                | 0.7 | 0.15 | 0.15 |

- a Find the expected percentage return on an investment.
- b Find how much an investment of \$50 000 is expected to be worth after 2 years.
- 2 A salesperson is starting work in a new region and analyses the probability of how many sales he is likely to make in the next month:

|                         |      |      |     |      |      |
|-------------------------|------|------|-----|------|------|
| Number of sales ( $x$ ) | 0    | 1    | 2   | 3    | 4    |
| $P(X = x)$              | 0.35 | 0.25 | 0.2 | 0.15 | 0.05 |

- a Given that he makes at least one sale, state the probability that he will make 2 sales.
- b The salesperson is offered 2 payment schemes:
- Option A: Flat monthly income of \$1800
  - Option B: \$1000 flat fee per month plus \$500 per sale
- If he chooses option B, what is his expected monthly income?
- c Which option should he choose to maximise his income?

- 3 At a car park in the city, all day parking is charged on the following basis below:

- Cars with just a driver pay \$25
- Cars with a driver and one passenger pay \$15
- Cars with a driver and at least two passengers pay \$12

The number of people in one of these cars on a given day is summarised in the table:

|                  |      |      |      |     |     |
|------------------|------|------|------|-----|-----|
| Number of people | 1    | 2    | 3    | 4   | 5   |
| Number of cars   | 4500 | 3500 | 1100 | 600 | 300 |

- a Find the probability a randomly selected car is carrying 3 people.
- b Given that a car was carrying at least 2 people, find the probability it was carrying 4.
- c Let  $X$  represent the parking fee paid by a randomly selected car. Construct the probability distribution table for  $X$ .
- d Find the expected revenue per car in this car park.

e Find the standard deviation of the revenue per car in this car park.



4 To be allowed to leave class and go to lunch, Mrs Ammar gives her class a 3-question multiple choice quiz, with each question consisting of 4 possible answers. David hasn't listened to a word Mrs Ammar has said all lesson and will have to guess each of the three questions.

- a Find the probability David will guess none of the questions correct.
- b Find the probability David will guess all of the questions correctly.
- c Find the probability David will guess just one of the questions correctly.
- d Let  $X$  represent the number of questions David guesses correctly. Construct a probability distribution table for  $X$ .
- e State whether the table represents a discrete probability distribution. Explain your answer.

5 When a student completes a task set by their teacher on Spacemaths, the number of hints used is monitored by the system:

- The probability of using at least 1 hint is 0.6.
- The probability of using 2 hints is the same as using 3 hints.
- The probability of using 1 hint is the same as using 4 hints.
- At most students can use 4 hints.
- The probability they use 2 hints is half the probability that they use 0 hints.

- a Let  $X$  represent the number of hints they used. Construct a probability distribution table for  $X$ .
- b Find the expected number of hints a student will use.
- c Given that a student used at least 2 hints, find the probability they used 4 hints.

6 Jo and Ky are playing a game of cards that is either won or lost, there is no draw. The probability that Jo wins the first game is 0.6.

- If Jo wins a game, the probability he wins the next game is 0.7.
- If Jo loses a game, the probability that Ky wins the next game is 0.8.
- They keep playing until either Jo or Ky wins two games.

- a Construct a probability tree diagram of this situation.
- b Let  $X$  represent the number of games of cards played before someone wins two games. Construct a probability distribution table for  $X$ .
- c Find  $E(X)$ .
- d Given that Jo won, calculate the probability that 3 games were played.

7 A fair standard die is thrown. Let  $X$  be the number of dots on the uppermost face.



- a Construct the probability distribution table for  $X$ .
- b Is the discrete probability distribution uniform or non-uniform?
- c Find  $E(X)$ .
- d Find  $\text{Var}(X)$ .

8 Two normal dice are rolled and the sum of the numbers on the uppermost face recorded. Let  $Y$  represent the value of the sum of the two dice.

- a Complete the table for this discrete probability distribution:

| $y$        | 2              | 3 | 4              | 5 | 6 | 7              | 8 | 9 | 10 | 11 | 12             |
|------------|----------------|---|----------------|---|---|----------------|---|---|----|----|----------------|
| $P(Y = y)$ | $\frac{1}{36}$ |   | $\frac{3}{36}$ |   |   | $\frac{6}{36}$ |   |   |    |    | $\frac{1}{36}$ |

- b State the most likely sum to occur.
- c Describe the shape of the distribution.
- d Find the expected value.
- e Find the variance.

9 A die was manufactured such that an odd number is twice as likely to be rolled as an even number. Let  $X$  be the number the die lands on.

- a Construct a probability distribution table for  $X$ .
- b Find  $E(X)$ .
- c Find  $\text{Var}(X)$ .

10 A game is played in which a standard six-sided die is rolled. If it lands on a number other than 1, then the score is that number. If it lands on 1, then a second four-sided dice with numbers 3 to 6 is rolled and the number that die lands on is the score. Let  $X$  be the score of a player in this game.

- a Construct a probability distribution table for  $X$ .
- b Find  $E(X)$ .
- c Find  $\text{Var}(X)$ .



11 A six-sided die with numbers from 1 to 6 is weighted such that  $P(\text{prime number}) = 0.1$  and  $P(4) = P(6) = 0.3$ . Let  $X$  represent the possible outcomes from one roll of the dice.

a Construct the probability distribution table for  $X$ .

b Find the following:

i  $P(X < 3)$

ii  $P(X = 3 | X \leq 5)$

iii  $P(X < 3 | X < 5)$

iv  $P(X < 4 | X \geq 2)$

12 A fair standard die is thrown onto the ground and the number of visible odd-numbered faces (the faces which are not on the ground) is noted. Let  $Y$  be the number of visible odd-numbered faces.

a Construct the probability distribution for  $Y$ .

b Is this discrete probability distribution uniform or non-uniform?

13 A fair standard die is rolled and the number of dots on the visible faces (the faces which are not on the ground) is noted. Let  $W$  be the number of dots that can be seen on the visible faces.

a Construct the probability distribution table for  $W$ .

b Is the discrete probability distribution uniform or non-uniform?

14 A regular six-sided dice has a side length of 8 cm. The dice is rolled on the ground and the height above ground of the dot on the face with only a single dot is noted. Let  $H$  be the number of centimetres this single dot is above the ground.

a List the possible outcomes for  $H$ .

b Hence, construct the probability distribution table for  $X$ .

15 Two dice are rolled and the absolute value of the differences between the numbers appearing uppermost are recorded.

a Complete the sample space in the given table.

b Let  $X$  be defined as the absolute value of the difference between the two dice. Construct the probability distribution table for  $X$ .

c Find  $P(X < 3)$

d Find  $P(X \leq 4 | X \geq 2)$

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 0 |   |   | 3 |   |   |
| 2 | 1 |   |   |   |   |   |
| 3 |   |   |   |   | 2 |   |
| 4 |   |   |   |   |   |   |
| 5 | 4 |   | 2 |   |   |   |
| 6 |   |   |   |   |   |   |

- 16** Two dice are rolled and the difference between the largest number and smallest number is calculated. A player wins \$1 if the difference is 3, \$2 if the difference is 4, \$3 if the difference is 5 and \$0 otherwise.

- Complete the sample space in the given table.
- Let  $X$  be the winnings from one game. Construct a probability distribution table for  $X$ .
- Find the expected winnings.
- If it costs \$2 to play each game, find the player's expected return.

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 2 |   |   |   |
| 2 |   |   |   |   |   |   |
| 3 |   |   |   |   |   |   |
| 4 |   |   |   |   |   |   |
| 5 |   |   |   |   |   |   |
| 6 |   |   |   |   |   |   |

- 17** At a local fair, in a game that involves rolling a standard six-sided die, players can win a prize depending on what they roll. Each player must pay \$3 to play. The prizes are awarded as follows:

- The player wins \$3 if a 1, 3 or 5 is rolled.
- The player wins \$6 if a 4 or 6 is rolled.
- The player wins \$9 if a 2 is rolled.

- Let  $X$  be the prize received by the player. Construct a probability distribution table for  $X$ .
- Find the expected prize value.
- Find the standard deviation of the distribution.

- 18** At a fair, a games stall operator offers prizes worth \$1.50, \$2, \$1 and \$0.50 for one attempt at a particular game. The probabilities of winning these prizes are respectively 0.15, 0.01, 0.05 and 0.03.

- Find the probability of not winning a prize.
- If each game costs \$2, find the expected profit per game for the operator in dollars.
- If each game costs  $C$  and the games stall operator made a profit of \$595 from 500 games. Find  $C$ , the amount he likely charged per game in dollars.

- 19** The probability that a particular biased coin lands on tails is 0.7. Let  $X$  be the number of tails when the coin is tossed twice. Complete the given probability distribution table for  $X$ .

| $x$        | 0 | 1 | 2 |
|------------|---|---|---|
| $P(X = x)$ |   |   |   |

**20** In a game of two-up, a person called the “er” tosses two coins:

- If the coins land with two heads up, then the Spinner wins and the gamblers lose.
- If the coins land with two tails up, the Spinner loses and the gamblers win.
- If the coins land one head up and one tail up, the Spinner tosses the coins again and the gamblers break even.

- Construct a tree diagram to represent all possible outcomes of tossing two coins.
- If each gambler bets \$3, and can win \$3 per toss, construct a probability distribution table for the profit of the gambler for one game of two-up.

**21** An unfair coin is tossed. The chance of tails facing upwards after the toss is 30%.

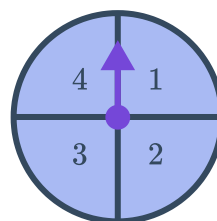
- Find the probability of the coin landing tails up for the first time on the third toss.
- Find the probability of the coin landing tails up for the first time on the fourth toss.
- Find the probability that it takes four tosses of the coin before you see a tail on the fifth toss.
- Let  $N$  be the number of tosses of the coin it takes before you see a tail on the next toss. Define the probability density function for  $N$ .

**22** Two fair spinners, A and B, are spun. The number from each spinner is noted and the total score is defined below:

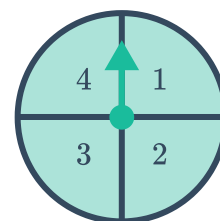
$$X = \begin{cases} A + B; & \text{if } A = B \\ A + B; & \text{if } A > B \\ B - A; & \text{if } A < B \end{cases}$$

- List all the possible outcomes of  $X$ .
- Construct a probability distribution for  $X$ .

Spinner A



Spinner B



**23** A spinner has four sections each numbered 1 to 4. The spinner is divided according to the given equations. Let  $X$  represent the number spun on the spinner.



- $P(1) = P(2) + P(3) + P(4)$
- $P(2) = 2P(3)$
- $P(3) = P(4)$

- a Construct the probability distribution table for  $X$ .
- b Hence, construct the cumulative probability distribution table for  $X$ .
- c Find the following:
  - i  $P(X < 3)$ .
  - ii  $P(X \geq 3)$ .
  - iii  $P(X = 1 \cup X = 3)$ .
  - iv  $P(X \leq 3 | X > 1)$ .

**24** Two spinners numbered from 0 to 4 are spun. Let  $X$  be the product of the two numbers that come up.

- a List all the possible outcomes of  $X$ .
- b Construct a probability distribution table for  $X$ .
- c Find  $E(X)$ .
- d Find  $\text{Var}(X)$ .

**25** Three marbles are randomly drawn from a bag containing seven black and three green marbles. Let  $X$  be the number of black marbles drawn.

- a Construct the probability distribution table for  $X$  if the marbles are drawn with replacement.
- b Construct the probability distribution table for  $X$  if the marbles are drawn without replacement.

**26** In Brad's toy box, there are 3 toy cars and 4 toy dinosaurs. Each day, for three days, he takes a toy at random and plays with it, and then puts it back.

- a Construct a probability tree diagram of all the possible combination of toys he could have played with over these three days.
- b Let  $X$  be the number of days he played with a toy car. Construct the probability distribution table for the discrete random variable  $X$ .
- c Find the expected number of days he will play with the toy car.
- d Find the standard deviation for the distribution of  $X$ .

**27** A pencil case contains 6 blue pens and 5 green pens. 4 pens are drawn randomly from the pencil case without replacement.



- Find the probability of drawing one blue pen from the pencil case.
- Find the probability of drawing three blue pens from the pencil case.
- Let  $X$  be the number of blue pens drawn. Complete the probability distribution table:

|            |                |   |                |   |   |
|------------|----------------|---|----------------|---|---|
| $x$        | 0              | 1 | 2              | 3 | 4 |
| $P(X = x)$ | $\frac{1}{66}$ |   | $\frac{5}{11}$ |   |   |

**28** A pencil case contains 9 blue pens and 5 green pens. 4 pens are drawn randomly from the pencil case, one at a time, each being replaced before the next one is drawn.

- Find the probability of drawing one blue pen from the pencil case.
- Find the probability of drawing three blue pens from the pencil case.
- Let  $X$  be the number of blue pens drawn. Complete the probability distribution table:

|            |                     |   |                      |   |   |
|------------|---------------------|---|----------------------|---|---|
| $x$        | 0                   | 1 | 2                    | 3 | 4 |
| $P(X = x)$ | $\frac{625}{38416}$ |   | $\frac{6075}{19208}$ |   |   |

**29** Two earrings are taken without replacement from a draw containing 3 black earrings and 5 brown earrings. Let  $X$  be the number of black earrings drawn.

- Construct a probability distribution for  $X$ .
- Given that at least one black earring was selected, find the probability that two were selected.

**30** A child randomly selects 3 balls without replacement from a box containing 8 red balls and 2 green balls. For every red ball chosen, the child receives 1 chocolate. For every green ball chosen, the child receives 5 chocolates.

- Let  $X$  be the number of green balls chosen. Construct the probability distribution table for  $X$ .
- Let  $T$  be the number of chocolates the child receives. Construct the probability distribution table for  $T$ .
- State the most likely number of chocolates that the child receives.
- State the expected number of chocolates that the child receives.